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Course: Artificial Intelligence

Experiment 7: Breadth First Search (BFS)

Aim: To implement the algorithm in Python.

Procedure:
1. Start the program.
2. Create a graph.
3. Traverse nodes using the algorithm.
4. Display the traversal order.
5. Stop.

Algorithm:
1. Initialize graph.
2. Visit source node.
3. Explore adjacent nodes.
4. Continue until all reachable nodes are visited.
5. Display output.

```
from collections import deque
graph={'A':['B','C'],'B':['D','E'],'C':['F'],'D':[],'E':[],'F':[]}
def bfs(start):
 visited={start};q=deque([start])
 while q:
  v=q.popleft();print(v,end=" ")
  for n in graph[v]:
   if n not in visited:
    visited.add(n);q.append(n)
bfs('A')
```

Observation: The traversal visits all reachable nodes correctly.

Result: The Python program was executed successfully.

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Experiment 8: Depth First Search (DFS)

Aim: To implement the algorithm in Python.

Procedure:
1. Start the program.
2. Create a graph.
3. Traverse nodes using the algorithm.
4. Display the traversal order.
5. Stop.

Algorithm:
1. Initialize graph.
2. Visit source node.
3. Explore adjacent nodes.
4. Continue until all reachable nodes are visited.
5. Display output.

```
graph={'A':['B','C'],'B':['D','E'],'C':['F'],'D':[],'E':[],'F':[]}
visited=set()
def dfs(v):
 visited.add(v);print(v,end=" ")
 for n in graph[v]:
  if n not in visited:
   dfs(n)
dfs('A')
```

Observation: The traversal visits all reachable nodes correctly.

Result: The Python program was executed successfully.